

SPECIAL ISSUE ARTICLE

On the Nature of Things

Island plant fire tolerance: Functional traits associated with novel disturbance regimes

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Oceanic islands (those that emerged from the seafloor) are incredibly diverse as a consequence of in situ diversification (Whittaker et al., 2017). These endemic island floras are celebrated for their unique biodiversity, which is increasingly threatened and endangered by anthropogenic activities and global change, including novel fire regimes (Fernández-Palacios et al., 2021). Wildfire regimes are highly variable among islands (Albrecht et al., 2016), with historically low frequencies on many remote, small islands as a consequence of there being few lightning storms (Trauernicht et al., 2024), leading to the prediction that island plants have converged on weak tolerance to regimes with intense, frequent fires (hereafter, fire tolerance) via evolutionary losses following colonization or failure to gain tolerance in situ (Figure 1, H1) (Burns, 2019). Fire tolerance of oceanic island plants is thus an urgent conservation question, and also scientifically intriguing for insights into plant functional diversity across islands.

Declines of oceanic island plants under modern fire regimes of escalated frequency and intensity seem to support the prediction that island plants lack fire tolerance (Fernández-Palacios et al., 2021), and certainly some of these declines are due to low fire tolerance. Yet, novel fire regimes coincide with other global change factors on islands including habitat loss, biological invasions, lost mutualists, and climate change, making it difficult to determine the role of fire in island plant declines. Moreover, ecosystem-scale disturbances such

as storms, volcanic activity, and drought have been features of many islands, hence, most island plants have likely evolved generalized disturbance tolerance strategies. Examination of the functional traits associated with fire responses provides the most direct test of island plant fire tolerance and is needed to investigate the resilience of island plants under novel fire regimes.

OCEANIC ISLAND DISTURBANCE REGIMES

Historical disturbance regimes are highly variable among oceanic islands (Fernández-Palacios et al., 2021). For example, volcanic eruptions can ignite natural wildfires (Nogales et al., 2022), and are likely to be sources of selection pressure for fire tolerance in some islands (Figure 1, H2). Although many oceanic islands have a volcanic origin and at least a history of fire (e.g., La Gomera, Kaua'i, Mauritius), with some still experiencing volcanic eruptions (e.g., La Palma, Hawai'i, La Réunion), others (e.g., uplifted islands such as Makatea Islands) have never had fires of volcanic origin. Proximity to continental source floras may also contribute to the occurrence of fire-tolerant species on oceanic islands, with dispersal to—and colonization by—fire-adapted species more likely for islands closer to continents (e.g., native *Pinus* species in Canary Islands and Isla Guadalupe; Figure 1).

Oceanic island floras are subjected to other frequent natural disturbances, which may select for tolerance strategies

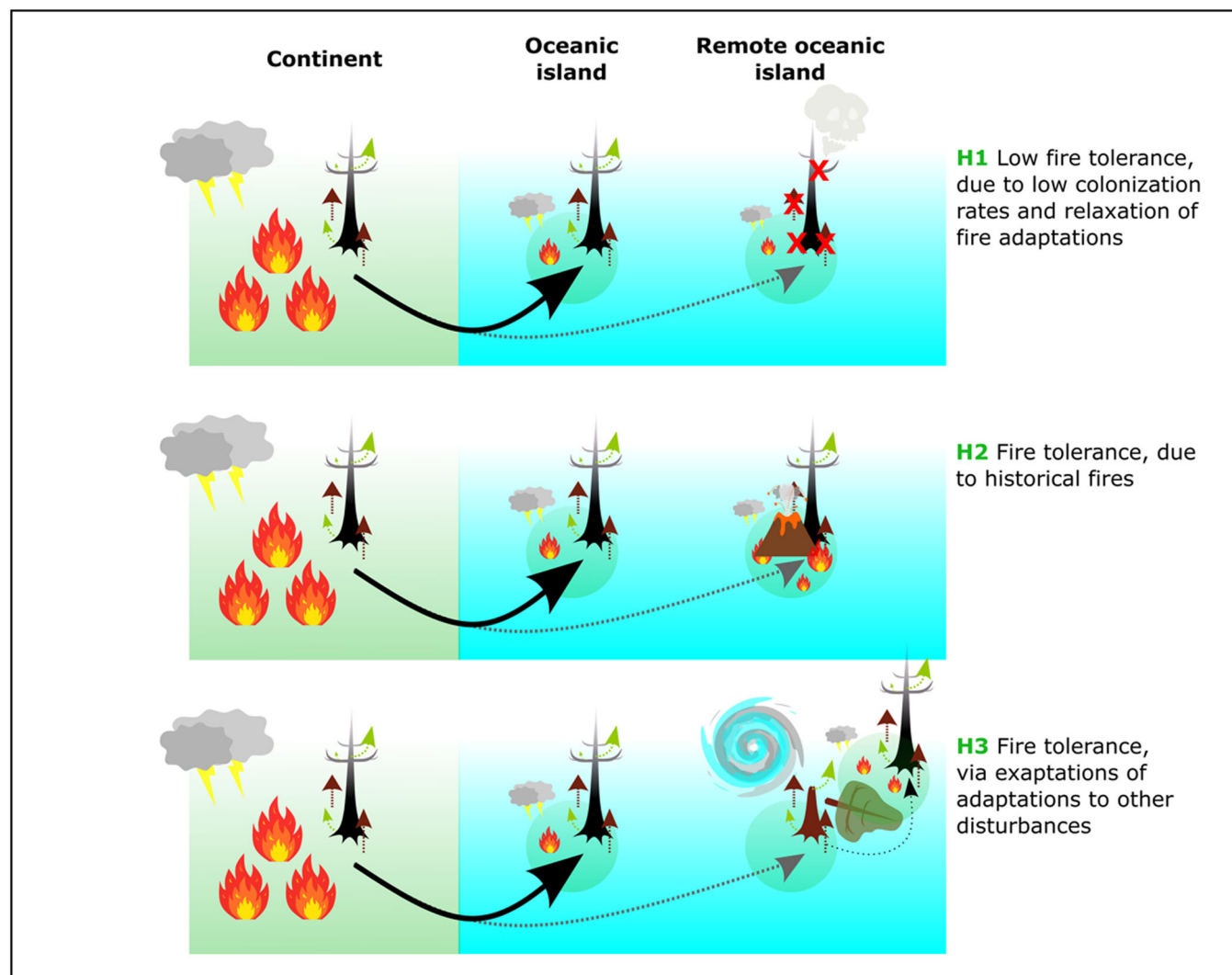


FIGURE 1 Model for trait-based tolerance to fire and other disturbances in oceanic island plants. Compared to continental floras, fire tolerance is expected to be lower in oceanic island plants as a consequence of weaker fire regimes and randomness in the dispersal of fire-adapted species, particularly to remote islands (H1). However, many oceanic islands with volcanic activity are expected to have plants with fire tolerance (H2). Furthermore, generalized disturbance tolerance traits may be present in oceanic island plants as a consequence of diverse disturbance regimes, and these traits may function in fire tolerance as exaptations (H3). Arrows depict dispersal from continents to oceanic islands, with rates reflected by arrow weight (solid—higher rates of dispersal; dashed—lower rates of dispersal). Green arrows from trees represent resprouting. In the absence of resprouting (red “x” marks), trees are killed by fire. Dotted brown arrows from the soil represent recruitment via the seeding strategy.

(Figure 1, H3). Storms and tropical cyclones, which break branches or trunks and uproot trees, frequently bring strong winds to many islands (Ibanez et al., 2022). Damage to island plants can also be caused by landslides, which are another common disturbance on islands caused by tectonic and storm events, as well as other climate extremes such as drought. Native island herbivores, especially insects, birds, and tortoises, can also cause widespread damage when they reach high population densities (Banko et al., 2022). Under these complex and interacting historical disturbance regimes, it is likely that island plants evolved disturbance tolerance strategies (Figure 1, H3), although direct tests of these interactions are challenging given the extreme shifts in island ecosystems under global change.

Oceanic islands are increasingly exposed to novel disturbance regimes, including fires of increased frequency and size

(Fernández-Palacios et al., 2021), and complex interactions among simultaneous global change factors are exacerbating novel fire regime effects on island species. For example, invasive plants can create high fuel loads, leading to fires that spread faster and farther than natural wildfires (Brooks et al., 2004). Climate change includes increasing temperatures and decreasing rainfall leading to conditions that promote fire (Kelly et al., 2020; Ibanez et al., 2022), while also constraining plant recovery following fire (Clarke et al., 2013). The consequence of these simultaneous global change factors is a novel fire regime of more frequent, hotter, and larger fires on islands (Trauernicht et al., 2024). Consideration of multifunctional tolerance strategies that are adaptive under these complex novel disturbance regimes on islands is urgently needed for mechanistic insights into island plant fire tolerance.

FIRE TOLERANCE IN OCEANIC ISLAND PLANTS

Resprouting and seeding (via fire-stimulated flowering, seed release, and germination, as well as dispersal into fire-disturbed areas) are two strategies that promote plant resilience to fire (Jaureguiberry and Díaz, 2023). Resprouting is a generalized disturbance tolerance trait, allowing plants to regrow from stem or root buds following damage to primary meristems (Clarke et al., 2013), usually as a consequence of damage from fire, herbivory (Archibald et al., 2019), or drought (Zeppel et al., 2015), and it is thought to be an ancient plant trait with a broad phylogenetic distribution (Pausas and Keeley, 2014). Resprouting has been clearly implicated in fire tolerance as illustrated by the high frequencies of resprouting species in fire-prone ecosystems (Pausas and Keeley, 2014); for these species, protection of buds from heat has also evolved via thick bark and location of buds belowground or sunken in stems (Clarke et al., 2013). Seeding is an evolutionarily derived fire tolerance strategy, involving the stimulation of flowering, seed release, or germination by fire-associated cues from smoke, chemicals, and heat (Pausas and Keeley, 2014), and flammability may coevolve with the seeding strategy because it promotes vegetation clearance that facilitates seedling establishment following fire (Jaureguiberry and Díaz, 2023).

The capacity to express resprouting and seeding strategies following fire has been investigated in few oceanic island plants. For example, in Hawai'i, 13 of 19 native woody and tree fern species burned by a lava-ignited wildfire resprouted from buds (Ainsworth and Kauffman, 2009). One of the best-known examples of fire-adapted island plants is the endemic Canary pine (*Pinus canariensis*), which resprouts from epicormic buds that are protected by thick bark (Arévalo et al., 2001). Fire resilience via seeding is less well documented, but has been reported in Samoa following wildfire (Hjerpe et al., 2001), and again for several native plants in Hawai'i (Ainsworth and Kauffman, 2009). These few examples highlight the lack of direct examination of fire tolerance traits in oceanic island plants. This is certainly a consequence of the hypothesis that oceanic island plants have low fire tolerance (Figure 1, H1). Yet, resprouting and seeding are adaptive to other disturbances in addition to fire, and it is challenging to confidently identify historical sources of selection pressure (Keeley et al., 2011). For example, many woody plants resprout following major storms, and the seeding strategy via high seedling recruitment has been detected in island plants following a category-5 cyclone on Tanna, Vanuatu (Ticktin et al., 2024). Thus, even if resprouting and seeding in island plants are exaptations to fire that evolved in response to other historical disturbances (e.g., storms, drought, herbivory), they are potentially adaptive under today's novel fire regimes (Figure 1, H3).

Characterization of resprouting and seeding strategies is an urgent research priority for oceanic island plants facing novel fire regimes. In addition to documentation of resprouting rates following fire, traits associated with the development and protection of bud banks in the stems, roots, and other

belowground storage organs should be examined (Clarke et al., 2013). It is generally unknown whether island plants have thick bark to protect stem buds, and because the evolution of resprouting in response to other disturbances is not likely to have also selected for thick bark, it may be that oceanic island plants are able to resprout but vulnerable to heat damage during fires. While the stimulation of flowering, fruiting, or seed release is generally assumed to be associated with high fire frequency and predictability (Jaureguiberry and Díaz, 2023), and thus, may not be common for island plants, the protection of seeds in the seed bank and by seed coat traits may be relatively common.

CONCLUSIONS

Island biodiversity faces diverse threats, leading to disproportionately high rates of species endangerment and extinction (Fernández-Palacios et al., 2021). Oceanic island floras are likely to demonstrate the full range of fire tolerance, from very intolerant to tolerant via resprouting and seeding strategies. However, responses to fire have yet to be examined directly in most oceanic island species, making it difficult to identify the precise role of novel fire regimes, among other simultaneous global threats, in island plant declines. Characterization of functional traits underlying island plant responses to fire offers a promising way forward for a mechanistic understanding of fire-plant dynamics, and to predict island plant responses to continued global change in the future. This characterization can be further informed by data on phylogeny, time since arrival on an island, and fire selection pressure in the range of origin. Identification of multifunctional traits that are adaptive under novel fire regimes and that also play a role in tolerating climate change and interactions with invasive species, will be particularly informative, contributing to the management of island biodiversity under complex novel disturbance regimes (Kelly et al., 2020). Deposition of fire tolerance trait data into trait-based restoration tools will make these readily available to resource managers, facilitating the integration of fire tolerance as part of island conservation.

AUTHOR CONTRIBUTIONS

K.E.B. conceived of the essay and led the writing. All authors contributed to the writing and editing. T.I. developed the figure.

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